VR-Based Surgical Training: Effects on Skill Proficiency and Retention

Department of Education, Communication and Learning, University of Gothenburg

PDA675 – Technology, Knowledge and Learning: An Introduction

October 29, 2026

Introduction

Virtual reality (VR) simulation is a modern day tool that allows practitioners to train surgical procedures in realistic virtual environments before operating on live subjects. As the VR environment poses no risk to actual patients, high risk procedures can be performed repeatedly with no safety or ethical concerns. VR also allows training to occur where access to real patients or a wide range of training scenarios is limited.

Emerging evidence suggests that VR training can produce meaningful improvements in trainees' technical performance. Kley et al. (2025) found that orthopedic residents who completed multiple VR simulation training sessions showed significant improvement in surgical performance metrics such as steps completed, procedure time and skill scores. Coxe et al. (2025) also found that VR-trained participants consistently outperformed those using conventional training methods, suggesting benefits for skill development, safety and trainee confidence. However, it remains unclear how well these benefits extend to skill proficiency and retention.

Given this uncertainty and VR's practical relevance to surgical education, the research question this literature review aims to explore is: *How does the use of VR simulation affect skill proficiency and retention among practitioners undergoing professional surgical training?*

Method

This literature review used a systematic search approach, detailed in the PRISMA flow diagram located in Appendix A. As the topic of this review covers multiple disciplines — computer science, education and medicine — the articles selected were found using Scopus, a database known for providing peer-reviewed academic literature across multiple fields. The following string was searched within the field of *Article title, Abstract and Keywords*:

("surgical practitioner*" OR "surgical trainee*" OR "surgeon*" OR “surger*”) AND ("virtual reality" OR "VR simulation") AND ("workplace*" OR "hospital*" OR "clinic*") AND ("skill retention" OR "skill proficiency")

Initial identification

The initial search resulted in 40 works. The following database filters were then applied, and results were excluded for not being articles (n = 4), not being published in English (n = 0), or publication dates before 2020 (n = 24). Only articles published after 2020 were included to ensure that the findings reflect the outcomes of modern systems rather than older, outdated ones. This left 12 articles for the screening stage.

Title and Abstract Screening

The purpose of this phase was to identify studies involving certified surgeons or residency-program trainees undergoing professional surgical training using VR technology (excluding augmented reality and desktop-only simulations), with a focus on skill proficiency and retention.

As it was difficult to find eligible studies targeting the exact population (see *Citation Search* below), this criterion was widened to include advanced medical students without prior clinical experience, provided the technology and focus were highly relevant.

From the 12 articles screened, several were removed for ineligible technology (n = 3), non-empirical studies (n = 5) and ineligible study focus (n = 1). This left 3 articles for eligibility assessment.

Full-text Screening

The remaining three articles were fully screened against the same criteria mentioned above. One final article was excluded due to ineligible study topic (n = 1) as it focused exclusively on the technical/engineering process of developing a VR simulator rather than evaluating training outcomes. This left 2 eligible articles for the review.

Citation Search

As the initial search resulted in fewer articles than anticipated, citation searching (snowballing) was conducted on the reference list of the article by Ju et al. (2025), one of the relevant literature reviews from the original search. This list consisted of 24 sources, with several removed due to ineligible technology (n = 8), publications before 2020 (n = 7), ineligible document type (n = 4) and ineligible study focus (n = 2). This left a remainder of 3 articles, which were then screened and deemed eligible. A total of 5 articles (2 from the initial search and 3 from snowballing) were included in this review.

Table 1.
Articles included in literature review.

Publication Year	Authors	Title	Journal	Volume	Issue	Page #s
2025	Kumar et al.	Innovating medical education using a cost effective and scalable VR platform with AI-Driven haptics	Scientific Reports	15	1	N/A
2023	Tronchot et al.	Virtual reality simulation training improve diagnostic knee arthroscopy and meniscectomy skills: a prospective transfer validity study	Journal of Experimental Orthopaedics	10	1	N/A
2024	Rahman et al.	Hip Arthroscopy Simulator Training With Immersive Virtual Reality Has Similar Effectiveness to Nonimmersive Virtual Reality	Arthroscopy - Journal of Arthroscopic and Related Surgery	40	12	2840-2849
2024	Vamadevan et al.	Variable practice is superior to self-directed training for laparoscopic simulator training: a randomized trail	Surgical Endoscopy	38	4	1902-1911
2021	Huri et al.	Cadaver versus simulator based arthroscopic training in shoulder surgery	Turkish Journal of Medical Sciences	51	3	1179-1190

Results

Most included studies were randomized controlled trials. Only Tronchot et al. (2023) employed a non-randomized comparative design due to geographical constraints. All relied on

quantitative performance metrics, validated skill-assessment scales (e.g., Arthroscopic Surgical Skill Evaluation Tool (ASSET), Objective Structured Assessment of Technical Skill (OSATS)) or platform-generated data, with Kumar et al. (2025) and Vamadevan et al. (2024) assessing retention following a period of inactivity. A thematic analysis was carried out on each study (see Appendix B) with the following themes identified:

VR vs. Traditional Training Methods

Several studies compared VR training directly against traditional training methods. Kumar et al. (2025) found that trainees using an AI-driven haptic VR platform showed a 38% reduction in training time, a 42% improvement in procedural accuracy, and a 45% decrease in error rates compared with a conventionally trained group. Huri et al. (2021) had more limited results, with most measured outcomes showing little to no difference between those who trained on VR and those who trained using cadaver-based methods. However, VR-trained participants completed the procedure significantly faster ($p < .05$) and caused significantly less humerus cartilage scratching ($p < .05$).

VR Platform and Training Design Choices

Two of the studies examined whether specific design choices within the VR training affected outcomes. Rahman et al. (2024) compared immersive VR (headset) against non-immersive VR (screen and manikin) for hip arthroscopy training. When assessed via OSATs and ASSET, scores showed no significant difference in technical proficiency between the two different platforms. Vamadevan et al. (2024) compared structured variable practice against self-directed practice for laparoscopic training, finding that the variable practice group reached proficiency much faster during both the basic skills phase ($p = .015$) and after 3-5 weeks of inactivity ($p < .001$ overall), even though both groups used identical VR tools and spent the same amount of time training.

Skill Transfer

Competence and skills acquired through the VR environments seem to successfully transfer to physical, real world surgical procedures. Tronchot et al. (2023) found that junior orthopaedic residents who had additional VR arthroscopy training scored significantly higher on ASSET when tested on cadaveric knees in comparison to those who received standard training alone. These findings were extended by Rahman et al. (2024), where both immersive and non-immersive VR-trained groups displayed similar ASSET and OSATS results when performing a procedure on cadaveric tissue. Vamadevan et al. (2024) conducted a transfer test entirely in the VR environment. Trainees in the structured variable practice group were able to apply the basic skills learned to a novel procedural task (a salpingectomy) not included in their initial training, reaching proficiency significantly faster than the self-directed group ($M = 103$ min vs. 183 min, $p < .001$). Unlike Tronchot et al. (2023) and Rahman et al. (2024) however, this transfer was never tested on physical tissue or in a clinical setting.

Discussion

The aim of this literature review was to identify how the use of VR simulation affects skill proficiency and retention among practitioners undergoing professional surgical training. The reviewed studies suggest VR simulation is largely associated with positive surgical training outcomes, though the strength of these benefits varies depending on the comparison made, the training's design choices, and the type and testing method of transfer.

VR vs. Traditional Training Methods

The results of Kumar et al. (2025) align well with Ertmer and Newby's (2013) discussion of the behaviourist framework, identifying the "use of reinforcement to impact performance [tangible rewards, informative feedback]" (p. 49) as a core principle. This is reflected in how the

study used AI-driven haptic feedback to give trainees immediate, in-scenario correction, reinforcing responses as they occurred. Such immediate correction is the process behaviourism predicts should strengthen performance, especially for low-level procedural skills like speed and motor accuracy, which may explain why the haptic feedback implemented by Kumar et al. (2025) produced consistent, positive results.

The findings of Huri et al. (2021), in contrast, did not show such clear differences. However, these findings should be viewed cautiously, as both groups were tested using the same VR device, which only the VR-trained group had practiced on. This is a design limitation the study's authors acknowledge, and may explain why VR's advantage was found in measures related to familiarity with the simulator interface (e.g., moving the virtual camera, knowing how sensitive the virtual tools were to avoid scratching). In contrast, measures less linked to the use of the VR simulation (e.g., overall total scores) showed no difference.

VR Platform and Training Design Choices

The findings regarding design choices suggest that *how* VR training is designed matters more than the technology used to implement it. The results closely align with Ertmer and Newby's (2013) discussion of the cognitivist framework, which emphasizes "structuring, organizing, and sequencing information to facilitate optimal processing" (p. 53) as a core principle of instructional design. This is shown in the structured variable practice condition designed by Vamadevan et al. (2024), where task order was randomized, but training still imposed a clear structure, ensuring every skill was practiced an equal number of times. This design choice reflects the cognitivist concept of schema building, where organizing mental structures in a particular way makes them easier to retrieve later. Since both groups in Vamadevan et al. (2024) used the same VR tool and received equal training time, the results suggest that training design affects proficiency and retention.

The findings of Rahman et al. (2024) — which showed no significant difference between immersive and non-immersive VR use — fit alongside the cognitivist framework. If learning outcomes are shaped by how practice is structured, then the use of immersive or non-immersive VR is of little consequence, since neither platform structured practice differently from the other. Together, these articles suggest that VR's value as a training tool lies less in the immersion level of the technology, and more in the instructional design decisions surrounding it.

Skill Transfer

The findings on skill transfer were studied in two related but different ways. First, transfer from VR to the real world, and second, transfer from basic skills to a novel procedure within the VR itself. The first of these findings is supported by both Tronchot et al. (2023) and Rahman et al. (2024), who found that surgical skills acquired through VR training successfully transferred to surgery on a cadaver. This aligns with Selwyn's (2022) discussion of Lave and Wenger's situated learning, which states that "learning is situated in the 'real-life' environment where the practice or activity takes place" (p. 89), meaning that learning and competence are tied to the specific context where they are practiced rather than as something abstract (e.g., reading rules on how to perform surgery versus physically carrying out the surgery). In situations where real-life participation is not possible, "many educators see technology-based environments as an ideal means of supporting in situ forms of socially 'augmented' learning" (Selwyn, 2022, p. 89). This accurately captures VR's role as it is hands-on, interactive, and closely replicates surgical conditions, allowing what is learned within the simulation to transfer into real-life.

Vamadevan et al. (2024) showed a related but different form of transfer when they found that structured variable practice supported transfer to an entirely untrained procedural task within the VR environment itself. This fits with the schema-building idea discussed above as the variable group was forced to practice all four tasks a certain number of times in an

unpredictable, randomized order, while the self-directed group could choose their own order and repeat tasks as they wished. The balanced, unpredictable practice may have built more flexible mental structures, allowing the variable group to apply their basic skills more successfully to an untrained task. However, unlike Tronchot et al. (2023) and Rahman et al. (2024), this transfer was only tested within the VR environment, so it is unclear whether it would extend to a clinical/real-world setting.

Limitations

The studies examined show that VR training generally produces positive outcomes with none showing any negative consequences. However, this review faces several limitations.

Only a single database (Scopus) was used, despite the multidisciplinary nature of the topic. The small number of included studies could also limit how applicable these findings are in general. The population criterion was also expanded to include medical students, introducing variation in professional experience. Finally, the five studies varied in VR platform, surgical procedure, and outcomes measured, which further limits how directly the findings can be compared.

Despite these limitations, the evidence points toward VR simulation offering measurable benefits for surgical skill proficiency, retention and transfer. However, further research implementing more standardized methodologies and populations is required to discover how strong and widely applicable these benefits are.

References

Coxe, F. R., Stauffer, T. P., & Ast, M. P. (2025). Virtual reality simulation in orthopedic surgery education improves immediate procedural skill and knowledge acquisition, but evidence on cost-effectiveness and skill retention remains lacking. *Current Reviews in Musculoskeletal Medicine, 18*(10), 363–378. https://doi.org/10.1007/s12178-025-09973-8

Huri, G., Gülşen, M. R., Karmiş, E. B., & Karagüven, D. (2021). Cadaver versus simulator based arthroscopic training in shoulder surgery. *Turkish Journal of Medical Sciences, 51*(3), 1179–1190. https://doi.org/10.3906/sag-2011-71

Ju, C., Yu, H., Hao, S., Qin, Y., Yao, K., Hao, Z., & Fang, Y. (2025). Comparison between virtual reality with bench-top simulation in the acquisition of arthroscopic skill: A meta-analysis of randomized controlled trials. *Journal of Orthopaedic Surgery, 33*(2). https://doi.org/10.1177/10225536251374369

Kley, V., Chelli, S., Avram, G., Amsler, F., Leica, A., Elsheikh, R., Hirschmann, M. T., & Koch, M. (2025). Virtual reality simulation improves performance metrics in total knee arthroplasty training: A single-centre longitudinal study. *Journal of Experimental Orthopaedics, 12*(4), Article e70538. https://doi.org/10.1002/jeo2.70538

Kumar, A., Saudagar, A. K. J., Kumar, A., Alkhrijah, Y. M., & Raja, L. (2025). Innovating medical education using a cost effective and scalable VR platform with AI-driven haptics.*Scientific Reports, 15*(1). https://doi.org/10.1038/s41598-025-10543-8

Rahman, O. F., Kunze, K. N., Yao, K., Kwiecien, S. Y., Ranawat, A. S., Banffy, M. B., Kelly, B.T., & Galano, G. J. (2024). Hip arthroscopy simulator training with immersive virtual

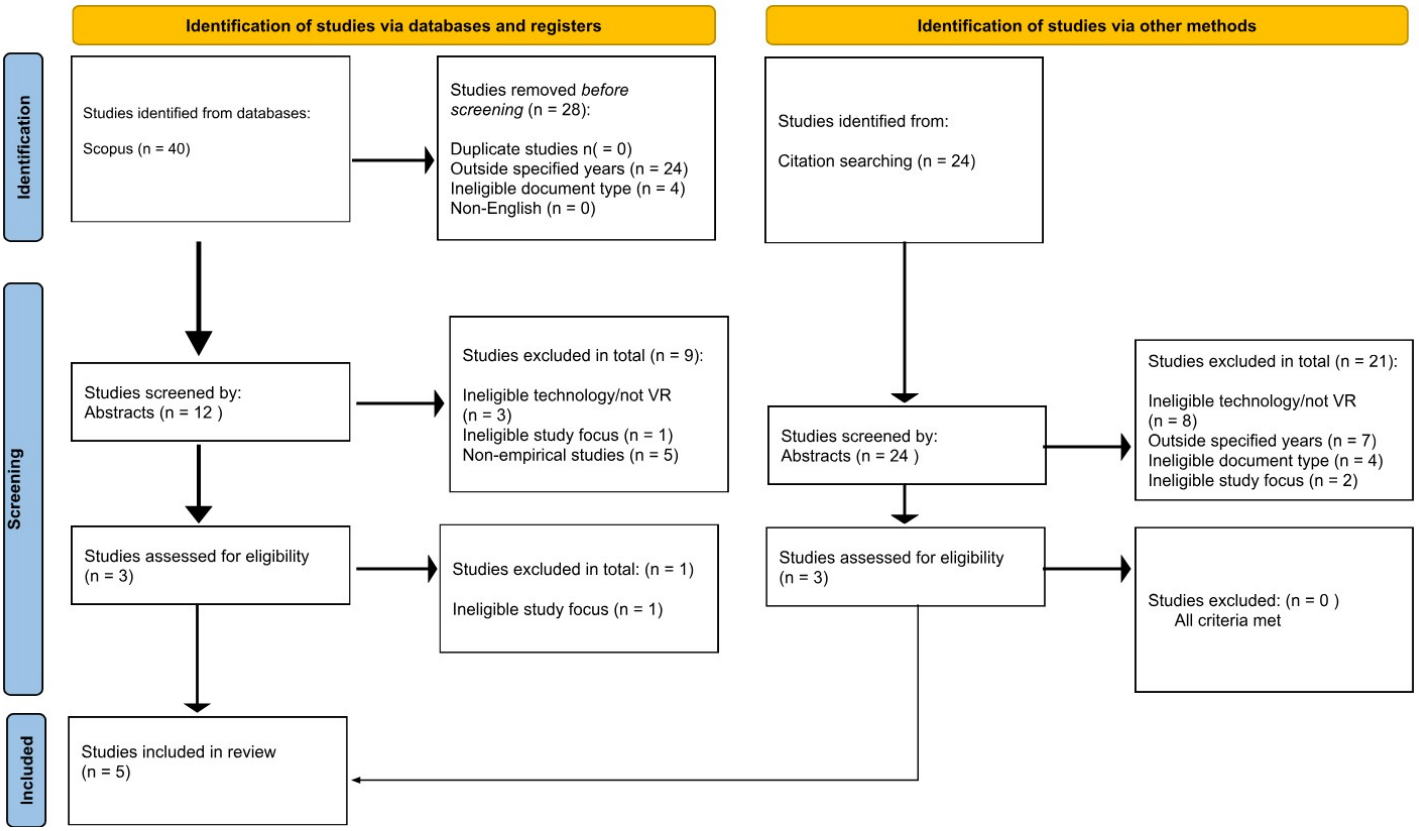
reality has similar effectiveness to nonimmersive virtual reality. *Arthroscopy, 40*(12), 2840-2849. https://doi.org/10.1016/j.arthro.2024.02.042

Tronchot, A., Casy, T., Vallee, N., Common, H., Thomazeau, H., Jannin, P., & Huaulmé, A. (2023). Virtual reality simulation training improve diagnostic knee arthroscopy and meniscectomy skills: A prospective transfer validity study. *Journal of Experimental Orthopaedics, 10*(1). https://doi.org/10.1186/s40634-023-00688-8

Vamadevan, A., Konge, L., & Bjerrum, F. (2024). Variable practice is superior to self-directed training for laparoscopic simulator training: A randomized trial. *Surgical Endoscopy, 38*(4), 1902–1911. https://doi.org/10.1007/s00464-024-10688-z

Appendix A

PRISMA Flow Diagram



Appendix B

Thematic Analysis Matrix

Year	Article Title	Purpose	Theoretical Basis	Methods	Skill Measured	Study Focus	Findings & Conclusion
2025	Innovating medical education using a cost effective and scalable VR platform with AI-Driven haptics.	To build and evaluate educational benefits of a VR training platform that uses AI-driven adaptive learning and haptic feedback in order to provide interactive and immersive instructions for wide range of medical procedures.	Not explicitly stated but terms adaptive learning and reinforcement learning modules are found in Fig.3. Platform gives immediate, in-scenario corrective feedback which is in line with behaviourist theory of learning (stimulus - response - reinforcement).	Randomized controlled trial (RCT) with participants of 87 graduate students, residents, professionals from multiple institutions. They were split into 2 groups: VR group: used platform for training in different techniques, e.g., laparoscopic surgical procedures, catheterization, emergency situations Standard training group: trained in traditional way, with physical simulators and clinical placements Data collection occurred over six months and groups performed identical procedures and assessments.	Procedural accuracy, time to completion, error rate, retention of learning skills, confidence gain	Proficiency and Retention: immediate and tested 3 months after training, but unclear what actual retention assessment method was used as it is not described in the paper. (VR vs. conventional training)	VR group significantly outperformed the non-VR group on procedural accuracy (42% improvement), training time (38% reduction), error rate (45% reduction), and trainee confidence (48% increase) and 90% skill retention reported at 3 month follow-up. VR platform allows personalized training, flexible, self-paced, repeatable practice that removes geographical, financial, and logistical constraints associated with traditional training (e.g., cadaver costs, instructor availability).
2023	Virtual reality simulation training improve diagnostic knee arthroscopy and meniscectomy skills: a prospective transfer validity study	To objectively investigate transfer validity in context of initial knee arthroscopy VR training.	Not explicitly stated but students are supervised on VR simulator performing same procedure repeatedly, then tested by doing same procedure on cadaver. This points to situated/sociocultural learning theory.	Participants were 36 junior resident orthopaedic surgeons with no prior arthroscopic experience. Since VR training and final evaluation were only done on site, the participants were divided into two non-randomized groups to account for geographical and travel time considerations: Non-VR: Received standard training VR: Received standard training plus 1 hour/month on VR surgery simulator.	Overall technical/procedural skill via ASSET total score.	Retention not assessed. Participants trained and tested after 6 month period of training. Proficiency measured through ASSET scores via transfer test of VR to cadaver. (VR-supplemented vs. conventional training + transfer to cadaver)	VR group scored significantly higher on the validated ASSET across 3 of 4 testing contexts (global, diagnostic arthroscopy, and meniscectomy on cadaver), though the bench-top meniscectomy difference did not reach statistical significance (p=0.077). ASSET was reported only as an overall score, the results of each skill dimension is unclear. Results confirmed transfer validity of skills acquired during VR simulator training to bench-top and cadaveric models. Authors

Year	Article Title	Purpose	Theoretical Basis	Methods	Skill Measured	Study Focus	Findings & Conclusion
				Took a questionnaire at the start to determine how much they knew. After the training time of 6 months, both groups performed 3 exercises that were evaluated independently by 2 blinded trainers: 1. arthroscopic partial meniscectomy on bench-top knee simulator (physical simulation/machine) 2. supervised diagnostic knee arthroscopy on a cadaveric knee (real physical tissue) 3. supervised knee partial meniscectomy on cadaveric knee (real physical tissue) Also did a second questionnaire about acquired expertise and feeling of progress.			conclude that VR simulator training could safely improve competencies ahead of operating-room performance and help standardize resident training while tracking progress over time.
2024	Hip Arthroscopy Simulator Training With Immersive Virtual Reality Has Similar Effectiveness to Nonimmersive Virtual Reality	To compare efficacy of immersive VR vs non-immersive VR training on procedural knowledge/skill acquisition and evaluate cost-effectiveness of each.	Not explicitly stated, could argue consistent with cognitivist beliefs learning is about structure/content of practice and feedback and they are testing this via immersive vs. non-immersive. Situated learning could also have influence due to being VR trained and cadaver tested.	Randomization of 14 orthopaedic surgery residents using Feinstein Biostatistics Randomization Management System into two groups: Group 1 used immersive VR simulator Group 2 used desktop based non immersive VR simulator. Group 1 completed modules a minimum of 3 times with 1 week of equipment access Group 2 had a single 2-hour scheduled session Performance was	Time to completion of surgery in simulation and on cadaver, participant experience, arthroscopic assessment by experts using OSATS and ASSET and transfer of skill and cost comparison	Retention unclear, article does not explicitly state time between training and testing but says after the training, they went to a surgical bioskills lab and performed the diagnostic hip arthroscopy on a cadaver. Reads as happening the same visit or soon after training completion. Proficiency tested via transfer test of VR trained to cadaver and	iVR participants completed their training session at an average of 172 seconds of simulation time compared with 310 seconds of simulation time in the non-iVR group. The iVR group was measured using a proprietary Precision Score, and the non-iVR group was measured using a proprietary Total Simulation Score. There was no significant difference in “overall VR score” between the iVR and non-iVR groups (77.1% [13.7], 72.7% [14.5]; P = 0.57). Time to completion was

Year	Article Title	Purpose	Theoretical Basis	Methods	Skill Measured	Study Focus	Findings & Conclusion
				evaluated by 4 board-certified orthopaedic surgeons using a blinded video review of surgery performed on physical cadaver and using Objective Structured Assessment of Technical Skills (OSATS) and ASSET scores.		evaluated using OSATS/ASSET scores. (iVR vs. non-iVR + cadaveric assessment)	recorded for the cadaveric procedure. The iVR group completed the cadaveric procedure in an average of 69 seconds compared with 52 seconds in the non-iVR group (P = 0.33; ICC, -20.0 to 54.3). Post-training performance metrics of iVR and non-iVR participants were similar, perceptions of improvement was observed for all participants regardless of group and there is about 132 fold cost difference of non-iVR to iVR.
2024	Variable practice is superior to self-directed training for laparoscopic simulator training: a randomized trial	To use VR simulation to investigate whether variable practice is better than self-directed training in terms of retention and transfer to a procedural task. Also looked at how variable practice affected the need for instructor feedback and the mental and cognitive load experienced during training.	Variable training as a part of contextual interference theory (Battig): struggling more during practice helps one learn better later. Cognitive load theory (Paas, Renkl & Sweller 2003; Leppink 2017; Van Merriënboer & Sweller 2010): optimize learning by respecting capacity of human mind. Desirable difficulties (Bjork & Bjork, 2011): learning task that requires a lot of effort, slows down immediate performance but significantly improves long term retention and knowledge transfer.	There were 36 medical students randomized into 2 groups: Group 1: Trained using structured variable practice Group 2: Autonomous self-directed training. Both trained on a laparoscopic VR simulator. After training to proficiency on the basic skills tasks, trainees immediately did a transfer test on procedural task (a salpingectomy). They then had 3-5 weeks of inactivity and returned to run retention tests where all participants practiced till proficiency again using self-directed method and did the procedural task. Max of one session of 2-h allowed per day.	Time to reach pre-defined proficiency (during retention for both basic skills and procedural task). For mental and cognitive load experienced during training, SIM-Task Load Index questionnaire used to assess perceived cognitive load for simulation training consisting of: mental demands, physical demands, temporal demands, frustration, task complexity, situational stress, distractions, perceptual strain and task control.	Retention was determined by comparing time to proficiency from first and last session (3-5 weeks inactivity) of both retention and transfer tests. Proficiency-based training design, measured time to proficiency as main outcome variable. (Structured vs. self-directed VR training)	Variable practice is superior to conventional self-directed training for a proficiency-based laparoscopy simulator training program. Variable practice reduced time to reach proficiency, resulted in higher transfer to a procedural task, and improved retention due to a stronger consolidation of skills compared with self-directed training. Both groups significantly reduced their time to reach proficiency during the retention test compared to during the intervention. The reduction in time to reach proficiency shows that proficiency-based training program is resistant to skills decay over a period of 3–5 weeks.

Year	Article Title	Purpose	Theoretical Basis	Methods	Skill Measured	Study Focus	Findings & Conclusion
					The questionnaire was done after first and last session.		
2021	Cadaver versus simulator based arthroscopic training in shoulder surgery	To objectively evaluate whether VR medical simulators can effectively replace traditional cadaver dissections for improving surgical technical skills/surgical training.	Not stated, more of a VR vs. cadaver comparison of outcomes.	Participants were 34 orthopaedic practitioners and were randomized into two groups: Group 1 trained on VR simulator Group 2 used physical cadaver dissections Trained for 20 minutes and then both did an objective proficiency test on VR simulator.	Three scores calculated composed of number of parts: 1. Total score: simulation overview and procedure time 2. Safety score: scratching of glenoid and humerus cartilage 3. Economy score: camera and grasper path length	No retention testing, only single, same day testing with no delay. Proficiency directly investigated through purpose of study - whether VR training can provide comparable proficiency to cadaver training. (VR vs. cadaver training + test on VR device)	(Group 1 = VR, Group 2 = cadaver): Procedure time: Cadaver group took longer than VR group → VR group was faster ($p < .05$) Humerus cartilage scratching: Cadaver group had more scratching than VR group → VR group had less tissue damage ($p < .05$) Everything else showed no significant difference between groups. Training on cadavers is not significantly better than training on VR.